

Experimental Characterisation of 3D Patterns of Harmonic Radar Tags for Insect Tracking

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Abstract—An experimental methodology is proposed to measure the three-dimensional re-radiation patterns for compact harmonic radar tags. A vector network analyser with frequency-offset capability is used to measure the re-radiated signal from an X-band harmonic tag (9.2 GHz incident, 18.4 GHz re-radiated). The configuration uses a single antenna with a circulator to separate incident and re-radiated signals. Coupling through the circulator is removed via a background subtraction technique resulting in a noise-floor of approximately -105 dBm. Measurements of the tag show a dipole-like pattern with a directivity of 5.3 dBi. Peak tag efficiency was estimated to be approximately -14 dB at an incident power level of -19 dBm.

Index Terms—Antenna measurements, harmonic tag, harmonic transponder, nonlinear tag.

I. INTRODUCTION

Insects are a vital part of the ecosystem and the worldwide decline in insect numbers, due to climate change and other human activity, has significant ramifications for food webs, agriculture, human health, and the global economy [1], [2]. Effective conservation of many insect species requires detailed knowledge and understanding of their behaviour in real-world environments. Central to this effort is accurately tracking the movement of insects, which can provide insights into long- and short-term behavioural shifts due to changes in the environment [2]. However, the small size (below 2–3 cm) and weight (below 100–200 mg) of many insect species limits the use of *traditional* animal tracking techniques, e.g., using video, or active battery-powered transponder tags [3], [4].

Harmonic radar offers an alternative to traditional tracking methods. In a harmonic radar system, the radar illuminates a passive tag at a frequency f_0 , while a nonlinear element embedded within the tag generates harmonic frequencies. By tuning the receiver to the harmonic frequency, the nonlinear response of the tag can be isolated from surrounding linear

clutter. In the simplest case, a harmonic tag consists of a single antenna connected to a Schottky diode. Harmonic tags do not need an energy source (e.g., a battery or solar cell) and can thus be miniaturised sufficiently for attachment to insects and such systems have been successfully used with a large number of different insect species [4], [5]. However, the characterisation of these tags can be challenging, particularly at higher frequencies. Often, tags are electrically small and lack a ground plane, making accurate re-radiation measurements difficult [6]–[8]. As a result, most existing experimental characterisation methods rely either on the best-case (e.g., broadside) performance or on 2D plane-cuts. However, the latter approach can introduce alignment errors when the tag is repeatedly re-mounted on a positioner [8]. The former is also potentially insufficient for insect tracking applications, since insect-mounted tags continuously change orientation during movement, particularly in flight, causing tag detectability to depend strongly on the spatial distribution of the re-radiated response. Consequently, characterisation based solely on peak conversion efficiency or limited 2D cuts may provide an incomplete representation of practical tracking performance.

This paper focuses on developing a robust experimental methodology for measuring the full three-dimensional re-radiation patterns of harmonic radar insect tracking tags operating in X-band (between 9–10 GHz incident and 18–20 GHz re-radiated). A single dual-polarised antenna, combined with a circulator, is used both to illuminate the tag and measure the reradiated response, helping reduce alignment errors associated with previous two-antenna measurement approaches. The proposed system additionally employs filtering and signal processing techniques to suppress harmonic leakage coupled through the circulator, and full three-dimensional co- and cross-polarised reradiation patterns are captured within a single measurement process [9]. To the best of our knowledge, this is the first experimentally validated methodology for full 3D characterisation of harmonic radar tags.

This work was partially supported by the Royal Society of New Zealand, under a Catalyst Seeding Grant 24-UOC-010-CSG, and the Dutch Research Council via NWO VENI Grant 19156.

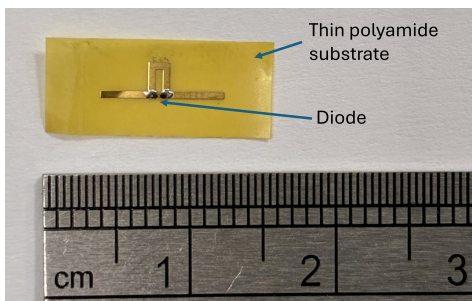


Fig. 1: Photograph of the harmonic radar tag considered.

II. HARMONIC RADAR AND TAGS

Harmonic radar is a nonlinear sensing technique that detects targets through the re-radiation of harmonics generated by nonlinear electronic elements embedded in a tag. The radar transmitter illuminates a scene with a continuous-wave or pulsed signal at a fundamental frequency f_0 . Tags containing nonlinear components, such as Schottky diodes or semiconductor junctions [6], generate signals at integer multiples of the incident frequency. By tuning the radar receiver to the harmonic frequency, most commonly the second harmonic $2f_0$, it can discriminate the nonlinear response from linear environmental clutter. As naturally occurring harmonic scatterers are rare and weak, harmonic radar provides high contrast and excellent target selectivity in complex environments.

Fig. 1 shows a photograph of the harmonic radar tag considered in this paper, which is a modified version of a tag from [8]. The tag consists of a printed dipole antenna with a Schottky diode connected at the feed point. An inductive loop is incorporated to provide impedance matching between the antenna and the nonlinear element. The tag is fabricated on a 25 μm -thick polyamide substrate and uses a MACOM Schottky diode (MA4E2502L) as the nonlinear element. The compact size (approximately $0.12\lambda \times 0.25\lambda$ at the 9.2 GHz incident frequency) and absence of a ground plane make accurate experimental characterization of the tag response particularly challenging.

III. EXPERIMENTAL METHODOLOGY

A. Single Antenna Configuration

Fig. 2 shows a block diagram of the experimental configuration. The signals are generated and measured using a vector network analyser (Keysight N5242A PNA-X), with frequency-offset capability, i.e., Port 1 transmits a 9.2 GHz continuous wave signal, and Ports 2 and 3 receive at 18.4 GHz. The network analyser is calibrated and set up to measure the received power (magnitude and phase), with a 10 Hz resolution bandwidth. A dual-polarised horn antenna (Flann DP240) is used to transmit and receive the signals, with a circulator (Pasternack PE8404) to separate them. The co- and cross-polarised responses are measured simultaneously. High-pass filters (Minicircuits ZHFW-K1042+) are needed in the receive chains to avoid overloading the network analyser receivers. Fig. 3(a) shows the relative location of the transmit/receive

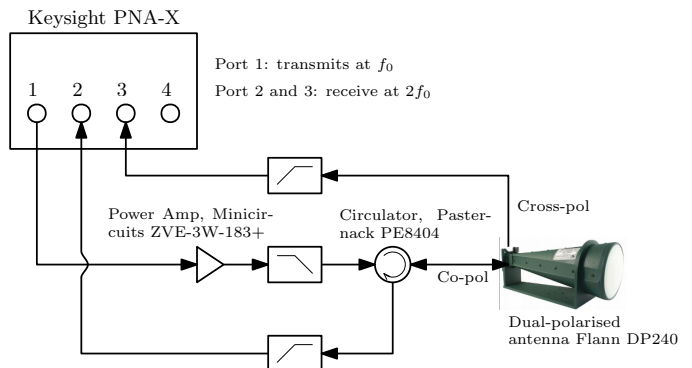


Fig. 2: Block diagram of the measurement setup. Not shown are the 2-axis positioner and tag mounting, depicted in Fig. 3.

horn (the network analyser is located behind the antenna mount) and the two-axis positioner; the separation distance is approximately 3.2 m. The angular resolution is 1° in elevation and 10° in azimuth, with trigger signals sent from the motor controller to the network analyser to ensure synchronisation.

B. Challenges

A significant challenge with measuring harmonic tags is ensuring sufficient RF power incident on the tag to produce the non-linear response. An external power amplifier (Minicircuits ZVE-3W-183+) was used to amplify the 9.2 GHz continuous wave signal from the network analyser to a maximum of approximately 32 dBm. However, at this level the 18.4 GHz second harmonic produced by the amplifier is approximately 15 dBm, which couples through the circulator and is also radiated through the horn. The signal coupling through the circulator can be removed digitally by subtracting the background response (i.e., when the tag is not present) in post-processing as described in Section III-C. Of greater concern is the radiation of the strong second harmonic component generated by the power amplifier, as this signal can scatter directly from the tag, positioner, and mounting structures, thereby corrupting the measured reradiated response. These effects are reduced by a low-pass filter (Minicircuits ZLSS-14G-S+) followed by a cavity resonant filter (1.5 dB insertion loss at 9.2 GHz and approximately 30 dB attenuation at 18.4 GHz).

Tag mounting on the positioner presents an additional challenge. Unlike in conventional antenna measurements, the harmonic tag is both electrically and physically very small and does not incorporate a ground plane, making the measured reradiated response particularly sensitive to nearby scattering objects. In practice, the metallic positioner and rotator structures can significantly perturb the measured pattern if insufficiently isolated from the tag. Several mounting configurations were therefore investigated. Mounting the tag at the tip of an RF absorbing cone, as shown in Fig. 3(b), was found to sufficiently shield the azimuth rotator while reducing the ground-plane effects that would otherwise be introduced by mounting the tag above a flat absorber. Additionally, the tag was oriented such that the expected null in the reradiation pattern was directed toward the rotator.

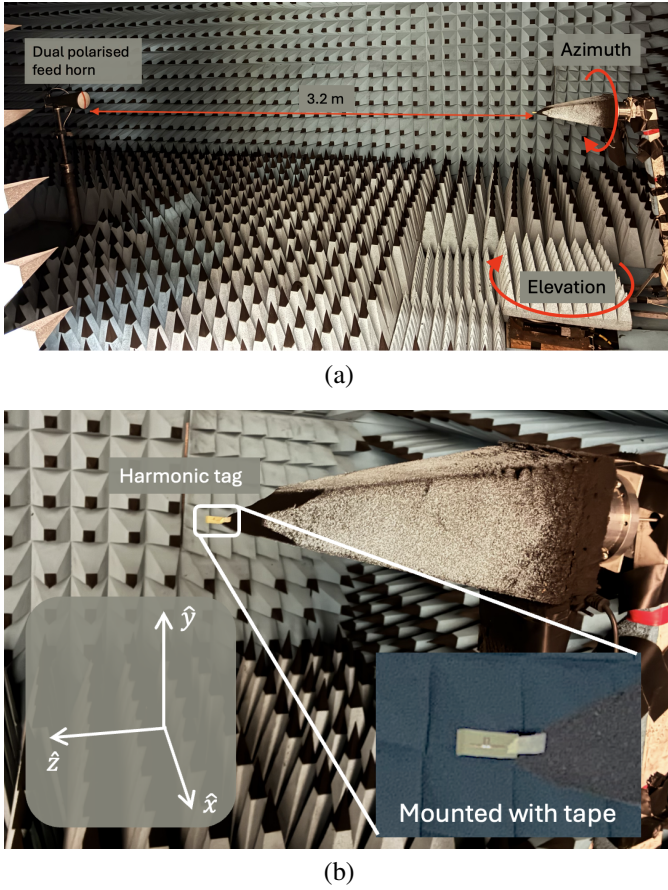


Fig. 3: Photographs showing: (a) the relative positions of the feed horn and the two-axis positioner and (b) tag mounted on the end of an absorbing cone and the corresponding coordinate system axes.

C. Filtering and Background Subtraction

At high transmit power levels, the second harmonic from the power amplifier is coupled through the circulator and is at approximately the same level as the response produced by the harmonic tag itself. As shown in Fig. 4, the received power therefore exhibits a floor of approximately -84 dBm. The ripples superimposed on this floor correspond to the actual response from the tag combined with the coupled second harmonic leakage from the power amplifier. The coupled component was observed to remain approximately constant and can therefore be removed through background subtraction, i.e., by performing a measurement without the tag present. Background measurements were performed both before and after the tag measurement to ensure consistency. Fig. 4 shows the received power after background subtraction. The resulting noise floor is approximately -105 dBm, and the re-radiation pattern of the tag becomes clearly visible.

IV. 3D ANGULAR RESPONSE

A. Effective Directivity of the Measured Harmonic Response

The background-subtracted measurements provide the received second-harmonic power as a function of tag orientation.

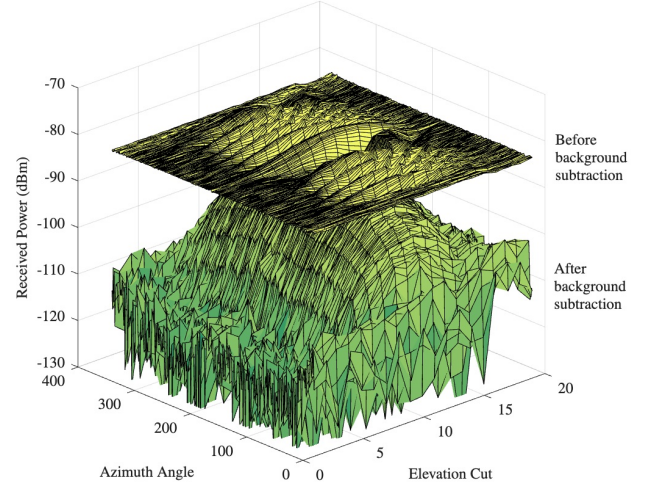


Fig. 4: Recorded power before/after background subtraction.

The co- and cross-polarised components (in linear units) are denoted by $P_{r,co}(\theta, \phi)$ and $P_{r,cross}(\theta, \phi)$, respectively, where θ is the polar angle measured from the tag dipole axis and ϕ is the azimuthal angle around this axis. The coordinate system is illustrated in Fig. 3(b), with the z -axis aligned along the dipole. The total measured response is given by

$$P_{r,tot}(\theta, \phi) = P_{r,co}(\theta, \phi) + P_{r,cross}(\theta, \phi). \quad (1)$$

The corresponding total power over the sphere is

$$P_{\Omega} = \int_0^{2\pi} \int_0^{\pi} P_{r,tot}(\theta, \phi) \sin \theta d\theta d\phi, \quad (2)$$

where $d\Omega = \sin \theta d\theta d\phi$ is the differential solid angle. The effective directivity of the co- or cross-polarised responses is then defined as

$$D_{co/cross}(\theta, \phi) = \frac{4\pi P_{r,co/cross}(\theta, \phi)}{P_{\Omega}}. \quad (3)$$

Since any angle-independent scaling of the measured response appears in both the numerator and denominator of (3), it cancels during normalisation. Consequently, the effective directivity describes how the available harmonic response is distributed over tag orientation, rather than its absolute strength.

Fig. 5 shows the co- and cross-polarised effective directivity components obtained from the measurement at an incident power of -19.1 dBm. The co-polarised response exhibits an approximately toroidal form, with the principal nulls aligned with the dipole axis. This is consistent with the combined patterns of the dipole-based harmonic tag at two harmonically spaced frequencies. The maximum total effective directivity is approximately 5.3 dBi. The peak of the cross-polarised response is approximately 10 dB below that of the co-polarised response. The observed asymmetry may arise from the tag geometry, mounting tape, and residual scattering from the supporting structure.

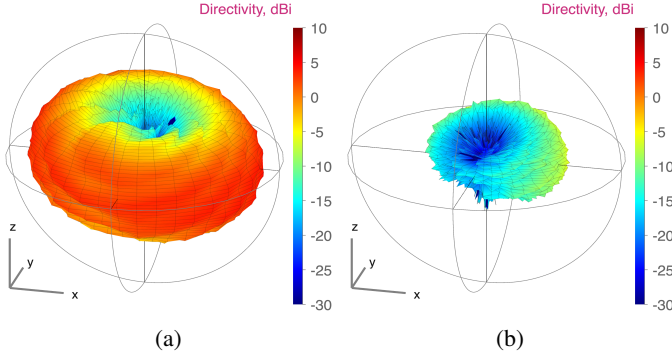


Fig. 5: Measured effective directivity of the harmonic tag shown in Fig. 1 at incident power level of -19.1 dBm: (a) the co-polarised and (b) the cross-polarised components. The patterns are normalised using the sphere-integrated total harmonic response according to (3).

B. Measured Conversion Efficiency Patterns

While the effective directivity characterises how the measured harmonic response is distributed over tag orientation, it does not quantify its absolute strength. The latter is described by the orientation-dependent tag conversion efficiency that is evaluated from received second-harmonic power $P_r(\theta, \phi)$ as

$$\eta(\theta, \phi) = P_r(\theta, \phi) - P_{\text{inc}} + L_{2f_0} - G_r + L_{\text{sys}}, \text{ dB} \quad (4)$$

where P_{inc} is the power incident at the tag, L_{2f_0} and G_r are the free-space path loss and the receive antenna gain at $2f_0$, respectively, and L_{sys} accounts for additional receiver-side system losses (see Table I). The effective directivity discussed above and $\eta(\theta, \phi)$ are directly related. Let

$$\bar{\eta} = \frac{1}{4\pi} \int_{\Omega} \eta(\theta, \phi) d\Omega \quad (5)$$

denote the spherical average of the conversion efficiency (in linear units). The corresponding effective directivity is $D(\theta, \phi) = \eta(\theta, \phi) / \bar{\eta}$. Therefore, the effective directivity determines the spatial distribution of the harmonic response, whereas the conversion efficiency sets its absolute level in the radar link budget.

Fig. 6 compares the measured co-polarised conversion efficiency cut at $\phi = 0^\circ$ for three incident power levels. The angular response remains broadly unchanged, while the peak conversion efficiency decreases by approximately the same amount (in dB) as the incident power, consistent with the expected quadratic operation of the tag at the low incident power levels [10]. At the lowest incident power, increased fluctuations are observed owing to the reduced signal-to-noise ratio and the limitations of the background-subtraction process. The maximum measured conversion efficiency is approximately -14 dB at $P_{\text{inc}} = -19.1$ dBm, -18.61 dB at $P_{\text{inc}} = -23.3$ dBm, and -22.97 dB at $P_{\text{inc}} = -28.4$ dBm.

V. CONCLUSIONS

This paper presented a methodology for experimentally characterising the 3D angular response of harmonic radar tags.

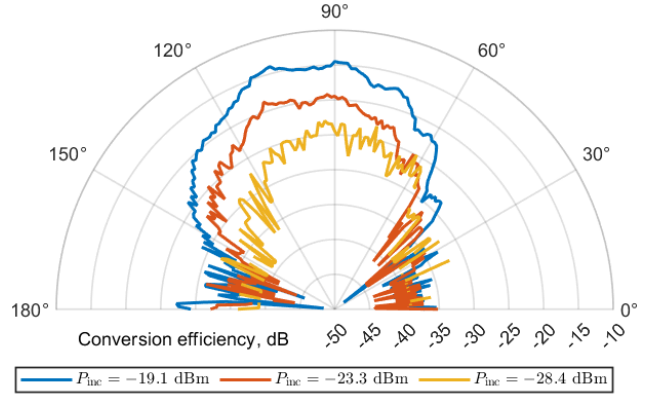


Fig. 6: Measured co-polarised conversion-efficiency cut at $\phi = 0^\circ$ for incident power levels of -19.1 , -23.3 , and -28.4 dBm.

TABLE I: Link-budget parameters.

Parameter	Symbol	Value
Receive antenna gain	G_r	17.5 dBi
Path loss at $2f_0$	L_{2f_0}	67.8 dB
System losses	L_{sys}	2.2 dB
Incident power	P_{inc}	$[-19.1, -23.3, -28.4]$ dBm

The proposed framework separates the spatial distribution of the tag response from its absolute magnitude, providing a more complete assessment of tag performance than peak conversion efficiency. The presented approach provides a practical basis for quantitative comparison and optimisation of harmonic tags.

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